

Baseline Model

```
seed(46)
DEBUG - Checking sample_weight:
[2.32066617 0.99185044 0.32742429 ... 0.32742429 0.65398742 2.32066617]
Model: "model"
```

Layer (type) Connected to	Output Shape	Param #
model_input (InputLayer)	[(None, 16, 20)]	0
norm_input (BatchNormaliza ['model_input[0][0]'] tion)	(None, 16, 20)	80
Conv1D_1 (QConv1D) ['norm_input[0][0]']	(None, 16, 30)	630
relu_1 (QActivation) ['Conv1D_1[0][0]']	(None, 16, 30)	0
Conv1D_2 (QConv1D) ['relu_1[0][0]']	(None, 16, 15)	465
relu_2 (QActivation) ['Conv1D_2[0][0]']	(None, 16, 15)	0
Conv1D_3 (QConv1D) ['relu_2[0][0]']	(None, 16, 10)	160
relu_3 (QActivation) ['Conv1D_3[0][0]']	(None, 16, 10)	0
act_pool (QActivation) ['relu_3[0][0]']	(None, 16, 10)	0
pool (GlobalAveragePooling ['act_pool[0][0]'] 1D)	(None, 10)	0
Dense_1_jetID (QDense)	(None, 32)	352

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['pool[0][0]']
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relu_1_jetID (QActivation) (None, 32) 0  
['Dense_1_jetID[0][0]']
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Dense_1_pT (QDense) (None, 16) 176  
['pool[0][0]']
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Dense_2_jetID (QDense) (None, 16) 528  
['relu_1_jetID[0][0]']
```

```
relu_1_pT (QActivation) (None, 16) 0  
['Dense_1_pT[0][0]']
```

```
relu_2_jetID (QActivation) (None, 16) 0  
['Dense_2_jetID[0][0]']
```

```
Dense_2_pT (QDense) (None, 8) 136  
['relu_1_pT[0][0]']
```

```
Dense_3_jetID (QDense) (None, 8) 136  
['relu_2_jetID[0][0]']
```

```
relu_2_pT (QActivation) (None, 8) 0  
['Dense_2_pT[0][0]']
```

```
relu_3_jetID (QActivation) (None, 8) 0  
['Dense_3_jetID[0][0]']
```

```
Dense_3_pT (QDense) (None, 4) 36  
['relu_2_pT[0][0]']
```

```
Dense_4_jetID (QDense) (None, 8) 72  
['relu_3_jetID[0][0]']
```

```
relu_3_pT (QActivation) (None, 4) 0  
['Dense_3_pT[0][0]']
```

```
jet_id_output (Activation) (None, 8) 0  
['Dense_4_jetID[0][0]']
```

```
pT_output (QDense) (None, 1) 5  
['relu_3_pT[0][0]']
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Total params: 2776 (10.84 KB)
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Trainable params: 2736 (10.69 KB)
Non-trainable params: 40 (160.00 Byte)

Epoch 10/10

153/153 - 4s -

loss: 1.3284 -

prune_low_magnitude_jet_id_output_loss: 1.2764 -

prune_low_magnitude_pT_output_loss: 0.0520 -

prune_low_magnitude_jet_id_output_categorical_accuracy: 0.4662 -

prune_low_magnitude_jet_id_output_weighted_categorical_accuracy: 0.4973

-

prune_low_magnitude_pT_output_mae: 0.2412 -

prune_low_magnitude_pT_output_mean_squared_error: 0.1095 -

prune_low_magnitude_pT_output_weighted_mae: 0.2282 -

prune_low_magnitude_pT_output_weighted_mean_squared_error: 0.1040 -

val_loss: 1.3126 -

val_prune_low_magnitude_jet_id_output_loss: 1.2625 -

val_prune_low_magnitude_pT_output_loss: 0.0501 -

val_prune_low_magnitude_jet_id_output_categorical_accuracy: 0.4799 -

val_prune_low_magnitude_jet_id_output_weighted_categorical_accuracy:

0.5037 -

val_prune_low_magnitude_pT_output_mae: 0.2411 -

val_prune_low_magnitude_pT_output_mean_squared_error: 0.1100 -

val_prune_low_magnitude_pT_output_weighted_mae: 0.2221 -

val_prune_low_magnitude_pT_output_weighted_mean_squared_error: 0.1004 -

lr: 0.0010 - 4s/epoch - 23ms/step



